Beyond U-Net: A Journey into something something DiT

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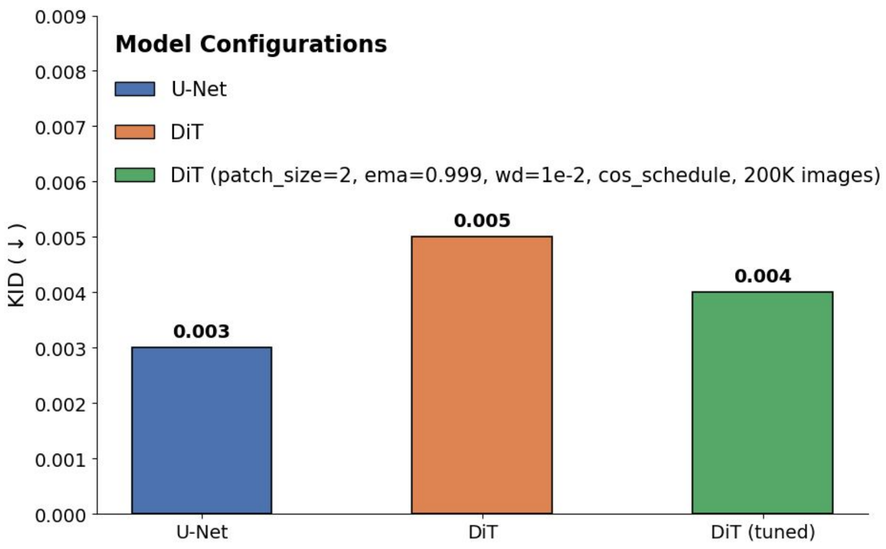
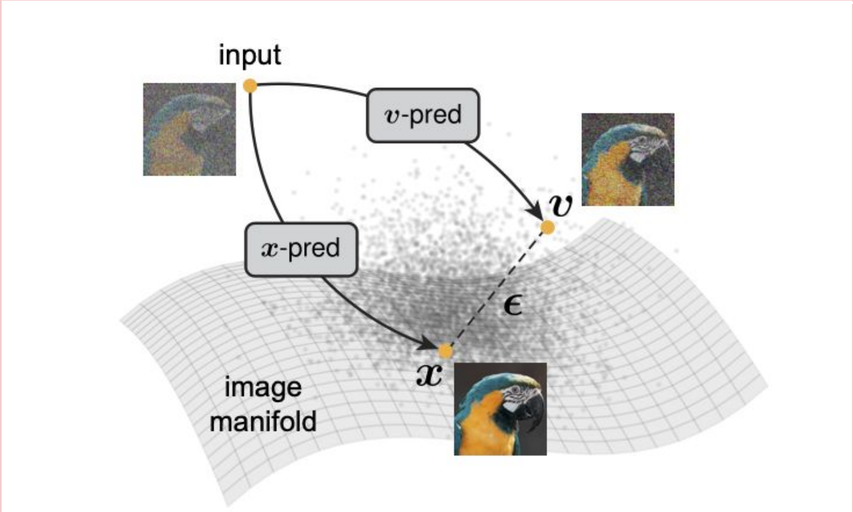


VAE Downsampling Mishap

Even ViT-S/1 can't bail us 🥲

Manifold Assumption to the rescue?



"The Manifold Assumption hypothesizes that natural images lie on a low-dimensional manifold within the high-dimensional pixel space"
- JiT (Li et al. 2025)

Algorithm 1 Training step

```
# net(z, t): JiT network
# x: training batch

t = sample_t()
e = randn_like(x)

z = t * x + (1 - t) * e
v = (x - z) / (1 - t)

x_pred = net(z, t)
v_pred = (x_pred - z) / (1 - t)

loss = l2_loss(v - v_pred)
```

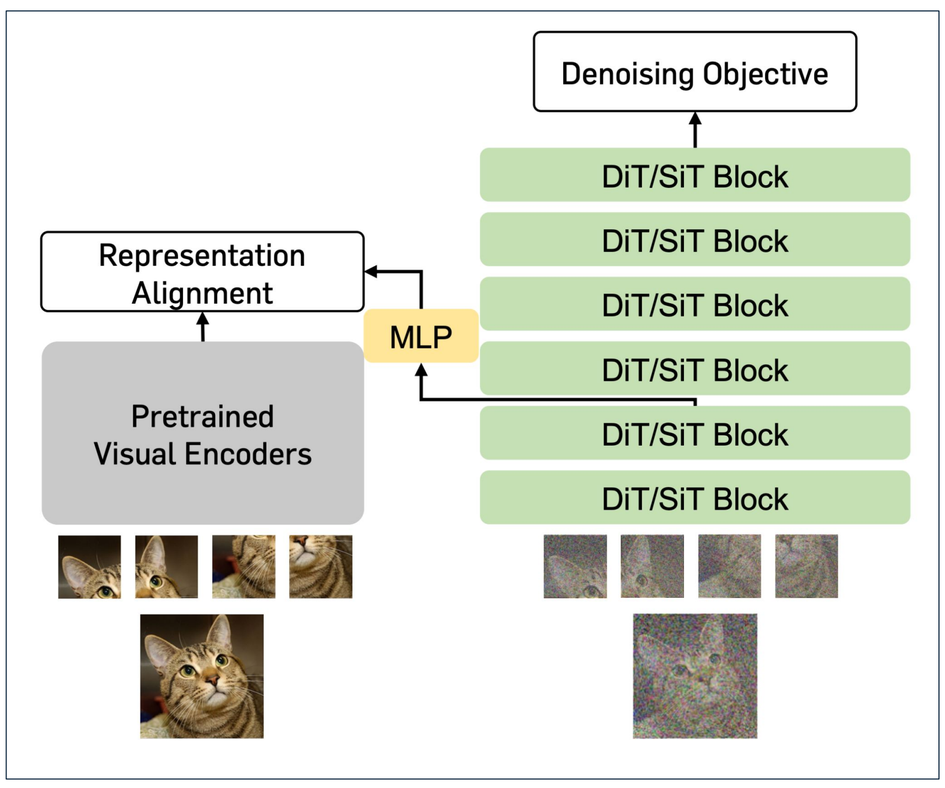
Algorithm 2 Sampling step (Euler)

```
# z: current samples at t

x_pred = net(z, t)
v_pred = (x_pred - z) / (1 - t)

z_next = z + (t_next - t) * v_pred
```

Learnings....

- U-Net's inductive goes a long way when data-bound.
- *Curse of Dimensionality* won't strike for patch size = 8 at 64x64 (each token 48 dim)
- Proportional Scaling of resolution required: 64 tokens can't represent a face 😉 (patch size = 8 at 64x64)

Representation Alignment



iREPA: 3 lines of code to make REPA better

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

Qualitative Results (DiT-S/8; 128x128)

v-pred

x-pred

x-pred + iRePA